UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS



Doctoral Program in Deep Tech, Artificial Intelligence & Emerging Technologies

Research Methods and Scientific Integrity in AI and Advanced Technologies

REPRODUCIBLE MACHINE LEARNING PIPELINE: SEED ROBUSTNESS, MLFLOW TRACKING AND DATA VERSIONING

DELIVERABLE No. 5

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1. Purpose and Deliverable Scope

This deliverable implements the reproducibility workflow requested in Session 5 of the doctoral course. The final rubric requires a Git history with at least six meaningful commits, a dataset versioned with DVC and pushed to a remote, two identical executions with the same seed, mean ± standard deviation over at least five seeds logged in MLflow, direct dependencies pinned with a Dockerfile that matches Python, and a README that allows a stranger to reproduce the reported number. The course frames reproducibility as a stack spanning code, data, experiments, environment, and reporting rather than as a single checkbox (Gaur, 2026).

I deliberately used the Wisconsin Diagnostic Breast Cancer dataset employed in the Session 5 workshop instead of inventing data for the doctoral AAC thesis. This choice isolates the methodological objective of Deliverable No. 5—reproducibility engineering—and produces an auditable template that can later be transferred to the doctoral dataset. The dataset contains 569 observations and 30 real-valued predictors derived from digitized fine-needle aspirate images (Wolberg et al., 1993).

The implementation goes beyond the minimum rubric in four ways: seven seeds are evaluated instead of five; six complementary performance metrics are reported; fold-level results are preserved as artifacts; and a machine-readable reproducibility manifest is generated in addition to the README, requirements file, Dockerfile, DVC metadata, MLflow database, results tables, and figures.

2. Rubric Alignment and Reproducibility Stack

Table 1

Deliverable No. 5 Rubric Alignment

Layer / Criterion	Evidence Generated	Status
Git history	7 D5 commits with clear messages	Exceeds 6+ requirement
Data	Dataset tracked by DVC + persistent Google Drive remote	Implemented in Colab workflow
Repeatability	Seed 42 executed twice; summary and all 5 folds must match exactly	Programmatic assertion
Experiment tracking	7 seeds logged to MLflow SQLite	Exceeds 5+ requirement
Reporting	Mean ± SD across seed-level means; 6 metrics	Implemented
Environment	Pinned direct dependencies + generated Dockerfile	Implemented
Stranger test	Fresh clone → dvc pull → run seed 42 → compare expected result	Included as final notebook gate

Note: Individual elaboration

The architecture follows the weakest-layer principle emphasized in class: perfect code versioning cannot compensate for an unversioned dataset, and exact package versions cannot rescue a leaky preprocessing pipeline. Consequently, the deliverable treats each layer as independently auditable.

3. Dataset and Experimental Design

Table 2

Dataset and Evaluation Protocol

Item	Specification
Dataset	Wisconsin Diagnostic Breast Cancer
Source	UCI / scikit-learn copy
Observations	569
Predictors	30
Positive class	Malignant = 1 (212 cases)
Negative class	Benign = 0 (357 cases)
Dataset fingerprint (MD5)	7d389a2808d8bcf9c30b5933557cdbad
Model	Logistic regression, liblinear solver
Preprocessing	StandardScaler inside Pipeline
Validation	5-fold stratified cross-validation, shuffled

Item	Specification
Seeds	7, 12, 21, 42, 101, 2026, 3407
Metrics	Accuracy, balanced accuracy, precision, recall, F1, ROC-AUC

Note: Individual elaboration

The seed controls the shuffled stratified fold assignment and the model random state. For every seed, the same five-fold protocol is used. The reported seed-level value is the mean of its five validation folds, while the within-seed standard deviation quantifies fold-to-fold variation. The headline result is then expressed as the mean ± sample standard deviation of the seven seed-level means. This design directly addresses the course warning that a single run gives no estimate of variance (Gaur, 2026; Pineau et al., 2021).

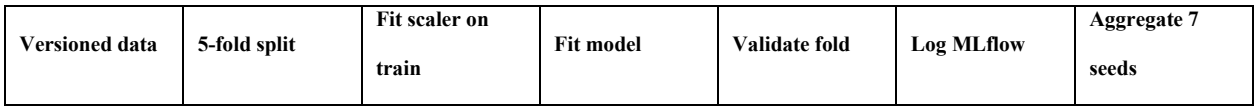
4. Leakage-Free Machine Learning Pipeline

The central methodological control is the use of a scikit-learn Pipeline containing StandardScaler followed by LogisticRegression. Because cross-validation calls fit separately inside each training fold, the scaler never computes its mean or standard deviation from validation-fold observations. This is the correct sequence: split within cross-validation first, fit data-dependent preprocessing on the training partition, and apply the learned transformation to the held-out partition.

This design prevents the preprocessing leakage demonstrated in Session 5, where scaling or feature selection before a split allows test information to influence model construction. Kapoor and Narayanan (2023) show that such leakage can produce systematically overoptimistic scientific claims. The deliverable therefore treats leakage control as a validity requirement, not merely an implementation detail.

Figure 1

Leakage-Free Reproducibility Workflow



Note. Author's own elaboration based on the Session 5 reproducibility stack.

5. Repeatability and Multi-Seed Results

5.1 Same-Seed Repeatability Test

Seed 42 was evaluated twice using the same data, pipeline, environment, and fold-generation logic. The complete result dictionaries were identical and the five fold accuracy values were bit-for-bit equal: 0.97368421, 0.95614035, 0.96491228, 0.99122807, and 0.99115044. This satisfies repeatability for the reference execution. It does not, by itself, prove validity or replicability; it only shows that the same team and setup can regenerate the same result.

Table 3

Same-Seed Repeatability Evidence

Execution	Seed	Accuracy mean	F1 mean	ROC-AUC mean
Run A	42	0.97542	0.96600	0.99538
Run B	42	0.97542	0.96600	0.99538
Comparison	—	Identical	Identical	Identical

Note: Individual elaboration

5.2 Seven-Seed Results

Table 4

Performance by Random Seed

Seed	Accuracy	Fold SD	Recall	F1	ROC-AUC
7	0.97716	0.01329	0.96213	0.96901	0.99504
12	0.98067	0.01143	0.96213	0.97360	0.99395
21	0.97895	0.01468	0.96246	0.97141	0.99539
42	0.97542	0.01567	0.94828	0.96600	0.99538
101	0.98067	0.01569	0.96678	0.97383	0.99554
2026	0.97718	0.01817	0.95759	0.96861	0.99511
3407	0.97887	0.01483	0.96224	0.97137	0.99561

Note: Individual elaboration

Table 5

Overall Stability Across Seven Seed-Level Means

Metric	Mean	SD	Minimum	Maximum	Range
Accuracy	0.97842	0.00195	0.97542	0.98067	0.52 pp
Balanced accuracy	0.97473	0.00263	0.96994	0.97781	0.79 pp
Precision (malignant)	0.98192	0.00323	0.97674	0.98626	0.95 pp
Recall / sensitivity	0.96023	0.00590	0.94828	0.96678	1.85 pp
F1 (malignant)	0.97055	0.00284	0.96600	0.97383	0.78 pp
ROC-AUC	0.99515	0.00057	0.99395	0.99561	0.17 pp

Note: Individual elaboration

Figure 2

Accuracy Stability Across Seven Random Seeds

Note. Error bars show the within-seed standard deviation across five stratified folds. The dashed line is the mean of the seven seed-level accuracy means.

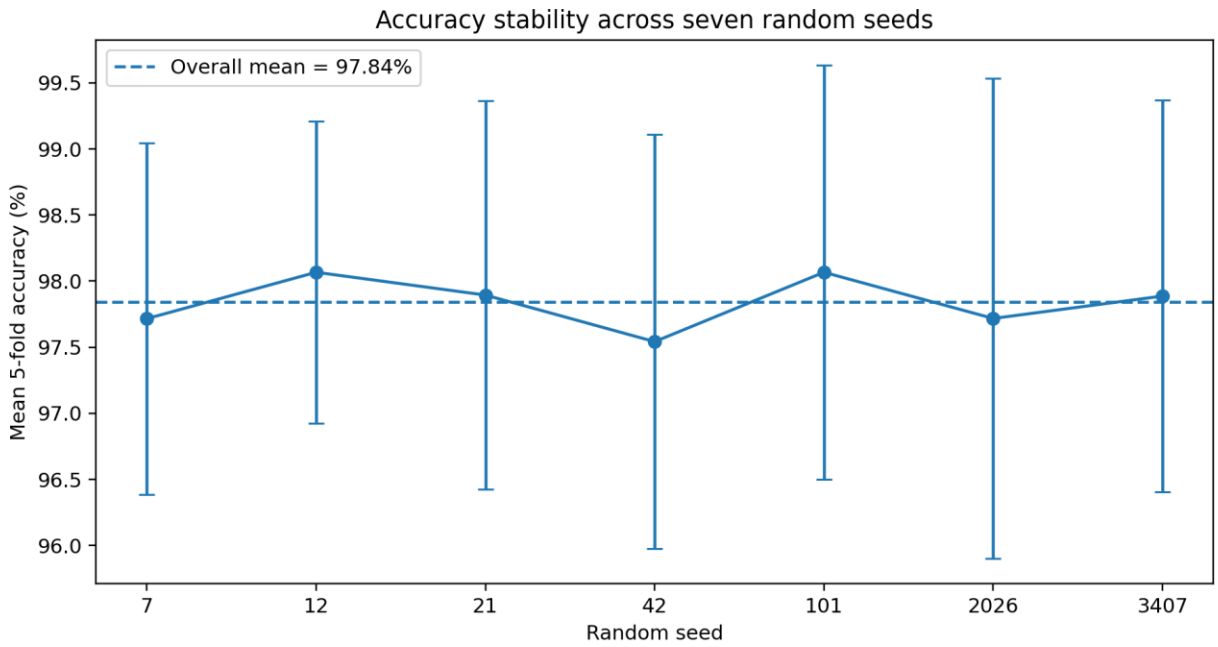


Figure 3

Multi-Metric Stability Across Seven Random Seeds

Note. Each point is the mean of five validation folds for the indicated seed.

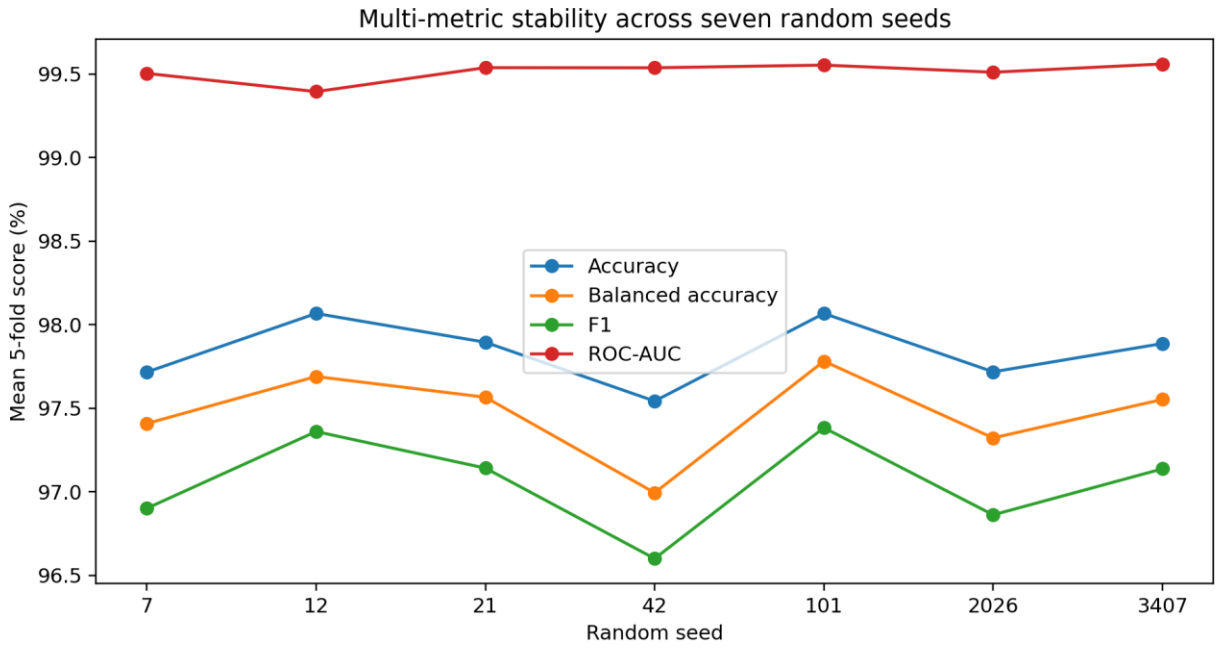
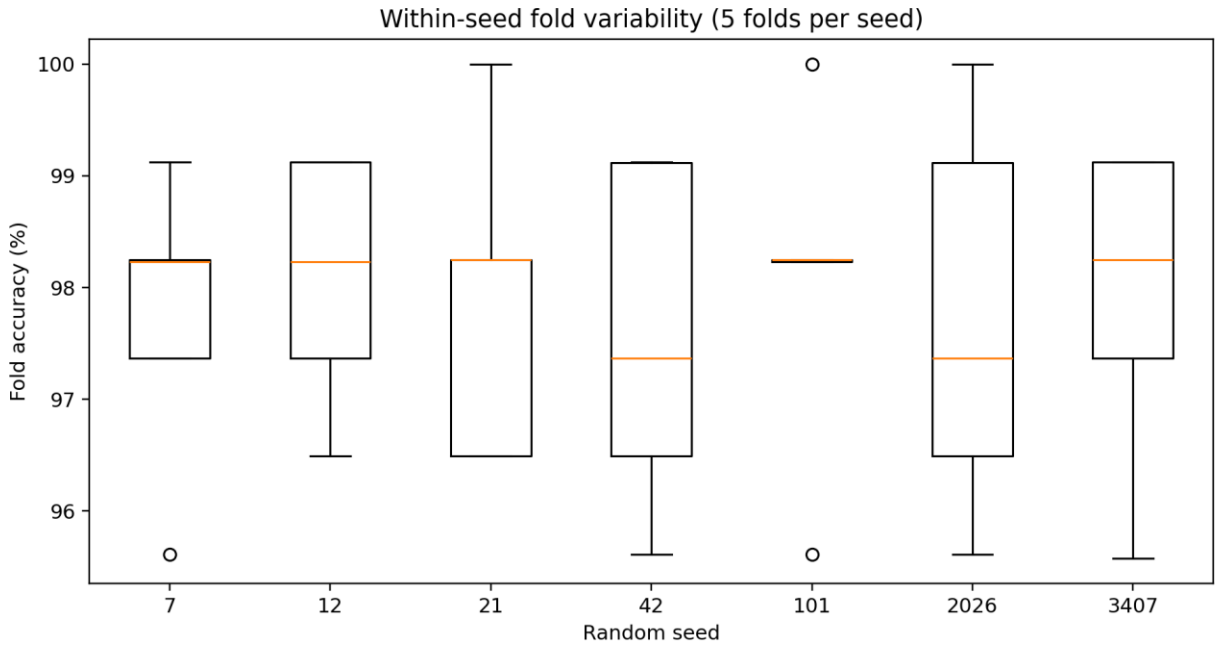


Figure 4

Within-Seed Fold Accuracy Distribution

Note. Each box summarizes the five fold-level accuracy values generated by one seed.



6. MLflow Experiment Tracking

MLflow is configured as the experiment ledger. Each of the seven seeds is recorded as a separate run with the seed, model specification, preprocessing definition, cross-validation protocol, current Git commit, dataset MD5 fingerprint, and all mean/SD metrics. Fold-level CSV files are logged as artifacts. The notebook uses a SQLite tracking URI rather than the legacy file store, consistent with the Session 5 material and current MLflow guidance for local database-backed tracking (MLflow Project, 2026).

Table 6

MLflow Metadata Logged Per Seed

Type	Logged content
Parameters	seed; model; preprocessing; CV specification; Git commit; data MD5

Type	Logged content
Metrics	accuracy mean/SD; balanced accuracy mean/SD; precision mean/SD; recall mean/SD; F1 mean/SD; ROC-AUC mean/SD
Artifacts	fold_metrics_seed_<seed>.csv
Backend	SQLite database: mlflow.db
Experiment	UNMSM-D5-reproducibility

Note: Individual elaboration

This design separates repeatability from robustness. A fixed seed permits an exact rerun, while the seven MLflow runs reveal how much the result changes when a legitimate source of randomness—the stratified fold assignment—changes. This is a more defensible scientific report than selecting the best seed.

7. Git, DVC, Environment and Stranger Test

7.1 Git History

The Colab creates seven Deliverable No. 5 commits, each corresponding to a meaningful methodological step. This makes the development process auditable and exceeds the final rubric requirement of six or more meaningful commits.

7.2 DVC Data Versioning

The raw CSV is tracked by DVC rather than committed directly to Git. The `.dvc` pointer and data `.gitignore` are versioned in Git, while `dvc push` sends the data object to a persistent Google Drive remote. The MD5 fingerprint provides a second independent check that the exact data file has not changed.

7.3 Environment Freezing

The notebook writes `requirements.txt` from the direct dependencies actually used by the workflow rather than applying `pip freeze` to the entire Colab environment. It then generates a Dockerfile whose base image matches the Python major/minor version of the runtime and fixes `PYTHONHASHSEED=0` at process start.

7.4 Stranger Test

After the GitHub push, the notebook performs a fresh clone into a new directory, retrieves the DVC-tracked data, runs the seed-42 command, and compares the resulting JSON with the expected value recorded in the README. This operationalizes reproducibility as "different execution context, same artifacts, same number," while acknowledging that a genuinely independent reviewer still needs read permission to the configured DVC remote.

8. Discussion

The principal result is not the high accuracy itself; it is the quantified stability of that result. Across seven seeds, accuracy was 97.842% ± 0.195% and varied by only 0.525 percentage points from minimum to maximum. ROC-AUC was even more stable, with a range of 0.165 percentage points.

However, malignant-class recall was more seed-sensitive: it ranged from 94.83% to 96.68%, a spread of 1.85 percentage points. This matters because in a diagnostic setting false negatives are substantively different from false positives. A single accuracy figure would conceal that clinically meaningful variability. The experiment therefore supports the course principle that seeds are necessary but not sufficient: researchers should report a distribution and should select metrics that reflect the scientific stakes of the problem.

The workflow also demonstrates why reproducibility is an engineering property of the whole research pipeline. The prediction code can be deterministic and still fail the stranger test if the dataset changes, dependencies drift, experiment metadata are lost, or instructions are incomplete. Git, DVC, MLflow, pinned dependencies, Docker, and the README address different failure modes and should be interpreted as complementary controls.

9. Conclusions

- The same-seed test passed: seed 42 reproduced identical summary and fold-level outputs in the reference execution.
- The leakage-free Pipeline ensures that scaling is learned only from training folds, preventing preprocessing leakage.
- Seven seeds show that the model is highly stable in accuracy and ROC-AUC, but recall is more sensitive to seed-dependent partitioning.
- The scientifically defensible headline is the mean ± SD across seeds, not a single best-seed result.
- MLflow creates an auditable experiment history; Git versions code; DVC versions data; requirements and Docker freeze the environment; and the README operationalizes the stranger test.
- The resulting Colab is reusable as the reproducibility scaffold for the doctoral AAC research once a real, ethically governed thesis dataset becomes available.

10. Limitations and Next Steps

This is a course reproducibility experiment on a public tabular dataset, not a clinical validation study. Cross-validation fold scores are not independent observations and should not be

treated as such for naive confidence intervals; the next course session addresses statistical comparison of models. In addition, the reference results were computed in Python 3.13.5 with scikit-learn 1.8.0; a different Colab environment can change last-digit values unless the pinned environment is recreated.

The DVC remote and GitHub push require credentials that cannot be embedded safely in a submitted notebook. The Colab therefore asks for user authorization at execution time and never stores the GitHub token. After the student executes those credential-dependent cells, the GitHub repository—not this Word report—becomes the authoritative reproducibility artifact.

References

Gaur, L. (2026). The reproducibility crisis in AI & ML [Course slides]. Research Methods and Scientific Integrity in AI and Advanced Technologies, Universidad Nacional Mayor de San Marcos.

Kapoor, S., & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. Patterns, 4(9), 100804. https://doi.org/10.1016/j.patter.2023.100804

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Pineau, J., Vincent-Lamarre, P., Sinha, K., Larivière, V., Beygelzimer, A., d'Alché-Buc, F., Fox, E., & Larochelle, H. (2021). Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program. Journal of Machine Learning Research, 22(164), 1–20.

Wolberg, W., Mangasarian, O., Street, N., & Street, W. (1993). Breast Cancer Wisconsin (Diagnostic) [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C5DW2B

Appendix A. Git Commit Plan

Commit	Message
D5-01	Initialize reproducible ML project structure
D5-02	Version dataset with DVC and configure remote
D5-03	Add seeded leakage-free evaluation pipeline
D5-04	Log seven-seed results and reproducibility figures
D5-05	Pin direct dependencies and add matching Dockerfile
D5-06	Document exact stranger-test reproduction recipe
D5-07	Add machine-readable reproducibility manifest

Appendix B. Exact Reproduction Commands

```
git clone https://github.com/mpatinoantonioli/UNMSM-Research-Methods-2026-II.git
cd UNMSM-Research-Methods-2026-II
dvc pull
pip install -r 05_Reproducible_ML_Pipeline/requirements.txt
PYTHONHASHSEED=0 python 05_Reproducible_ML_Pipeline/src/train.py --seed 42
```

Expected reference seed-42 accuracy mean: 0.97542. Expected dataset MD5: 7d389a2808d8bcf9c30b5933557cdbad. The authoritative output after final execution is the value written into the GitHub README by the Colab itself.